53 Analysis of data backup and recovery strategies in the cloud

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Abstract

This study examines modern data backup and recovery techniques in cloud computing settings. An in-depth literature analysis highlights the changing environment by examining the effects of various techniques. This study offers empirical insights into strategy choices and difficulties by employing a rigorous methodology that involves data collecting from diverse cloud service providers and enterprises. The results have demonstrated that choosing a cloud provider impacts how a plan is implemented and perennial concerns about data security, compliance, and cost management. Further, the latest technologies, including blockchain-based data integrity and artificial intelligence (AI)-driven anomaly detection, have also been discussed.

Keywords: Blockchain, data integrity, artificial intelligence, anomaly detection, diverse cloud service

Introduction

Modern information technology (IT) architecture must include cloud-based backup and recovery of data techniques. They entail securing digital data in cloud environments to guarantee data availability and integrity (Dalal, 2023). The flexibility and manageability benefits of cloud-based backup solutions enable businesses to preserve their data effectively. These tactics frequently make regular automated backups to keep data maintained and available in information loss, hardware problems, or emergencies. Organizations may quickly and effectively retrieve data with cloud-based recovery solutions, reducing delay and the risk of data loss. They provide a range of restoration options, including snapshot restoration, file-level recovery, and complete system recovery.

Further strengthening data resiliency, cloud-based recovery solutions frequently offer geographical and redundancy dispersion. In the current data-driven workplace, these measures are essential for protecting and safeguarding digital assets. These procedures guarantee the integrity, security, and accessibility of digital information stored in cloud settings. Companies are given the resources they need to effectively protect their priceless data assets due to their scalability and usability advantages. Automating frequent backups, which serve as a layer of protection against data loss, system problems, or unplanned occurrences, is one of the system's primary advantages. Data maintenance is greatly aided by cloud-based backup techniques, which allow businesses to quickly and effectively recover their data. This reduces downtime while lowering the danger of data loss, a significant concern in today's data-centric environment.

This study explores cloud backup and recovery procedures to balance accessibility, security, and cutting-edge technology successfully. This study offers empirical insights into strategy choices and difficulties by employing a rigorous methodology that involves data collecting from diverse cloud service providers and enterprises.

The remainder of the study has been organized as follows: The literature review, followed by elaborated methodology. The dataset is discussed, followed by the discussion of empirical results which finally concludes the study.

Related work

An expansion of specific applications within the communication system is made possible by the secure deployment of the Internet of Things (IoT) as an additional service within the information network (Dajun, 2021). Three primary components make up the IoT industry's development. The first component is recognition, which is a fundamental pre-condition. Second, it becomes clear that communication is a crucial platform and support. The ultimate goal that best captures the IoT's underlying purpose is application. High technological standards are required for the IoT's development and use, with solutions acting as a catalyst for advancement (Durga, 2022; Kiranpreet, 2022).

The fundamental architecture of the IoT is shown in Figure 53.1, which consists of a thorough application layer, av network infrastructure layer, a management

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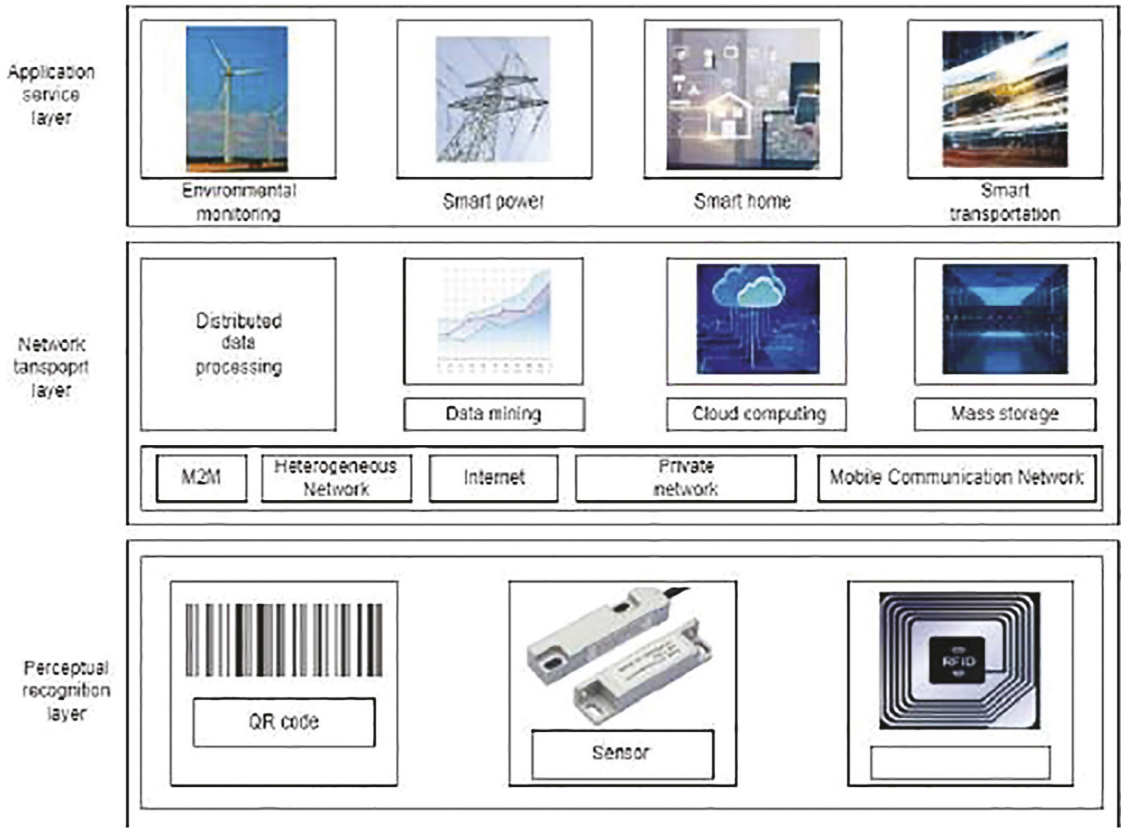

Figure 53.1 Fundamental IoT architecture

service layer, and a perceptual recognition layer. The central link bridging the material and informational worlds comprises the visual recognition layer powered by perception technology. This layer includes specialized adaptive electronic devices for human data retrieval and automatic data collecting tools, e.g., radio frequency identification (RFID) and sensor networks.

The network infrastructure layer's primary responsibility is to connect the Internet with lower-layer evaluation and recognition tools, providing access to applications at the upper layers. The Internet and the next generation form the basis of the IoT, supported by various wireless networks that provide Internet access and rely on robust computation and mass storage capacities for significant data collection.

The extensive application layer reflects the changing environment of online applications, which is influenced by the development of processing power (Iwona, 2023). Early data services focused on email and file transfers, but modern user-centric network applications include social networking, video streaming, and online gaming. Despite its increasing popularity, the IoT has inherent weaknesses caused by the enormous number of endpoints and difficulties associated with connectivity and collaboration among IoT devices, which are frequently sensitive to network attacks (Dalal, 2023). Focusing on Spark is a quick and comprehensive framework for handling massive amounts of data. Spark performs at speeds that are 100 times faster than Hadoop and MapReduce when memory resources are abundant. It outperforms these rivals by a factor of 10 when leaking data to disk, even when memory is limited. Spark's capabilities for complex directed acyclic graphs (DAGs), intended for in-memory data processing, give it its performance prowess. Spark is an object-based and operational programming framework implemented in Scala, allowing for the fluid manipulation of remote datasets akin to local collection objects (Brar et al., 2022; Rahul, 2022). Its defining characteristics are rapid implementation, user-friendly operation, adaptability, and compatibility (Lai, 2022).

A significant question is handling the difficult data backup task (Ramesh, 2023). Previously, companies or integrators were in charge of building these systems and ensuring they complied with all specifications and performed at their best. The IT environment in businesses has changed over time. Still, backup procedures frequently experienced alterations without a systematic methodology, omitting to consider the importance of syncing with basic standards

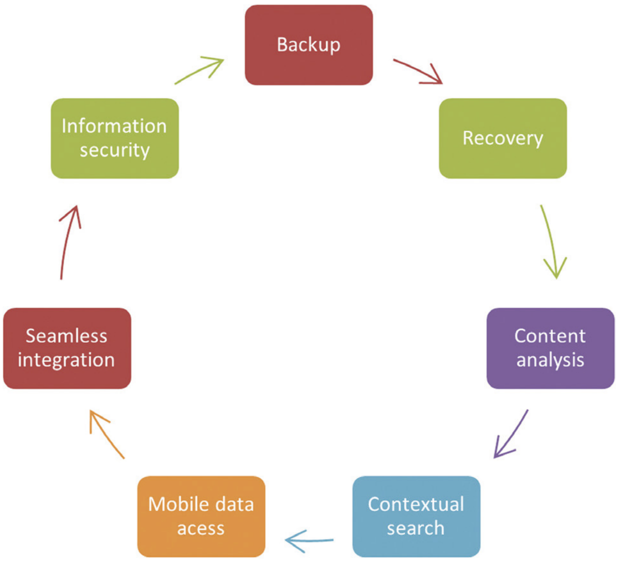

Figure 53.2 Important blocks of the proposed model

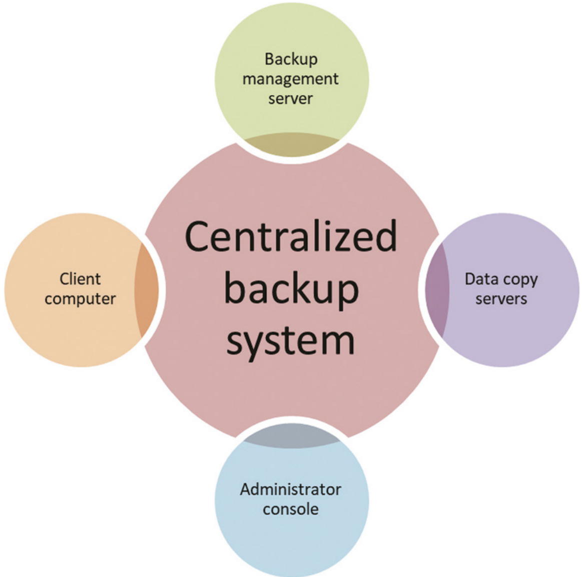

Figure 53.3 Components of the centralized backup system

throughout future phases. Understanding that a data safety system is only one part of a more extensive data security approach when developing one within a business is crucial. A system admin is responsible for keeping track of backup client PCs and recording hardware and storage devices while planning backup operations. The recovery management server has a specialized database keeping all relevant data. To backup data in line with chosen rules, schedules, or operator directions, the management server sends commands to agent applications installed on client PCs. These agent programs collect and send the data intended for backups to the copy server designated by the management server.

Several techniques are emerging to address concerns with data backup and recovery. The need to separate the quickness of these procedures from the amount of data is first and foremost. This can be done with the help of several tools that data storage devices, application programs, and resource management systems (RMS) vendors recommend (Rehman, 2022). For instance, snapshots allow for quick data backup and restoration with little performance impact and are frequently included in a larger policy framework. Second, the emphasis is on making it possible to recover particular data segments, eliminating the requirement for complete data restoration. Making functioning copies of production systems is made more accessible by solutions like "Oracle Standby", "DB2 HADR", and "MS SQL" constantly on, ensuring a speedy recovery from failure. Thirdly, there is a focus on closing the gap between the creation of data and its protection. Snapshots can be used as restore points to shorten backup intervals for less critical systems. Still, continuous data protection methods like "Oracle Standby with Flashback" may be required for more important systems.

Finally, it is essential to mitigate hidden flaws, which can be done by evaluating the efficacy of backups through restoration attempts. This procedure can be sped up by implementing efficiently recoverable computer instances, such as backup and standby systems. Some systems also have automated testing capabilities, allowing for recurring checks of data extraction, application accessibility, consistency, and reactivity. Different blocks of the proposed model have been shown in Figure 53.2 which includes backup, recovery, content analysis, contextual search, mobile data access, seamless integration, and information security.

In this situation, a centralized backup system includes a multi-layered design, as shown in Figure 53.3. It consists of client PCs with backup agent software. This backup management server can also perform data copy server duties, one or more data clone servers linked to backup devices, a backup system administration console, and data copy servers.

Methodology

Probability and data loss (PDL)

The probability of data loss can be calculated for data loss in the cloud from Equation 1.

$$PDL = 1 - (1 - R)^{N} \tag{1}$$

R denotes the reliability of a single backup copy, and N represents the number of backup copies.

For example, let R = 7 and N = 4, then PDL = 1 - (1-7) ^ 4 = -1295. This implies that PDL is extremely low, or it can be said that it is effectively zero.

Backups created from snapshots

A snapshot of the cloud's resources is taken using snapshot technology. It is a productive backup method without affecting system performance (Twana, 2022). Snapshots can retrieve information and are particularly useful when you need to return to a certain condition quickly. Vendors provide snapshot services, including AWS, Azure, and Google Cloud.

Replication and redundancy

Implementing data redundancy and replication across many cloud servers or zones of availability improves data availability and durability (Surbhi, 2015). This method stores data in several places to protect against calamities or outages at data centers. This is made more accessible by services like "Azure Geo-Replication" and "AWS S3 Cross-Region Replication".

Cloud-to-cloud restoration

Cloud-to-cloud backup methods can benefit businesses employing various cloud-based resources, such as SaaS apps. You can back up data using these services, such as "Veeam" or "Druva," from one cloud environment (such as Microsoft Office 365 or G Suite) to another cloud (such as "AWS, Azure, or Google Cloud"). For the protection of crucial corporate data kept on numerous cloud platforms, it is crucial.

Cost-benefit analysis

It draws insights into profitability when organizations shift to cloud computing in each layer. The three layers are base cost estimation, data pattern-based, and project-specific cost estimation. Equation 2 can be used to calculate the cost-benefit ratio.

$$\text{Cost} - \text{benefit Ratio} = \frac{\text{Value of Data-Cost of Backup and Recovery}}{\text{Cost of Backup and Recovery}} \quad (2)$$

Description of dataset

To get knowledge about this crucial field of IT, secondary data collecting for studying cloud-based backup and recovery of data solutions entails exploring current sources. Researchers can access various information from research databases, publications, and articles through a thorough literature review. Whitepapers, reports, and analyses that offer valuable data and trends are frequently found on sector-specific websites and in the documentation of cloud service providers (Surbhi, 2015). Additionally, market research studies and industry publications show valuable data and insights. Official government publications and regulations provide an additional source of reliable information. The most recent techniques and advancements in protecting data in the cloud can be found in the documentation provided by cloud service providers and on technology news websites. Online discussions, social networking sites, and specific groups encourage debate and user experiences, adding practical insights to the research. Books, conference proceedings, and seminars are all excellent resources for learning how recovering and backing up data in the cloud is changing (Xiaojun, 2022). Researchers can develop a comprehensive picture of the best practices, difficulties, and emerging trends in this vital topic by utilizing these secondary data sources.

The secondary data included in the Scopus authoring project was taken from various journal articles. It consists of a broad range of data drawn from these publications that have been carefully collected and arranged into Excel files. Many useful visualizations and graphic representations have been created using these Excel sheets. Data collection provides insightful information on various study problems and is the basis for the study's analytic approach.

Empirical results

The performance of data restoration and backup procedures in the cloud can be better understood through empirical data. These findings offer quantifiable information on restoration speed, recovery time, affordability, and dependability. They empower businesses to make wise judgments, improve their methods for cloud-based data security, and guarantee data resilience in changing cloud settings.

Figure 53.4 represents the challenge-response-verification process time overhead. It is clear that the processing overhead for challenge-response verification gradually increases as the number of challenge data blocks increases. Even if it exists, the rise in the process of verification overhead is still barely noticeable. Usually, the time cost of generating problems is far lower than that of generating responses. However, this cost gradually increases as more challenging data blocks are added. However, when the number of data blocks exceeds a critical threshold, such as 2000, the challenge's time cost significantly increases and converges with the confirmation process's cost.

Choosing proper algorithms is crucial for building an effective processing system for keeping data and backups within the Spark platform. The "APCA segmentation", "ratio R", "differential D", and "durationik T" techniques are all considered in this analysis.

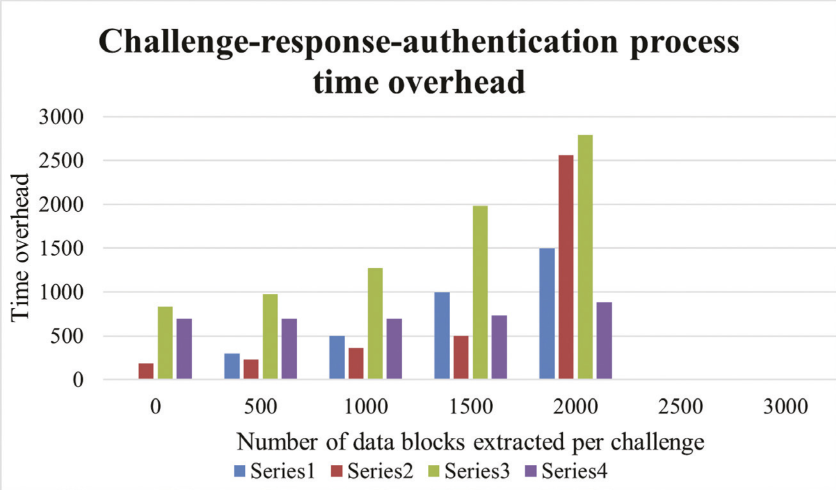

Figure 53.4 Challenge-response-authentication process time overhead

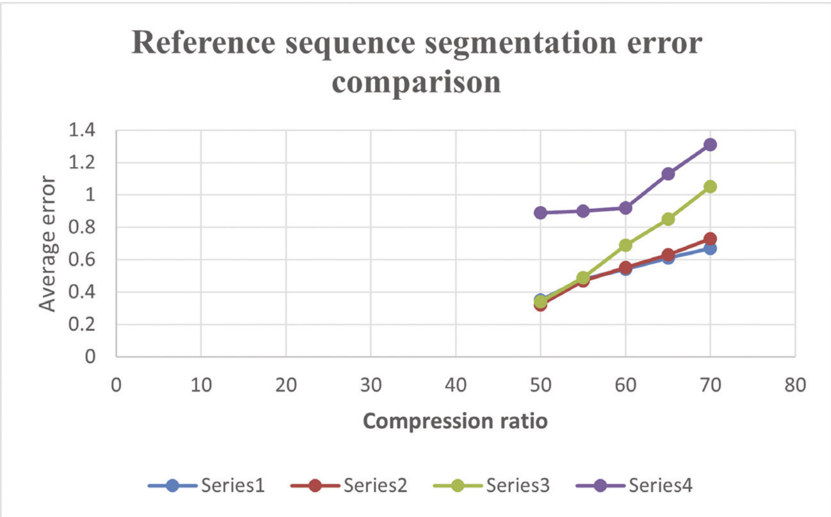

Figure 53.5 Reference sequence segmentation error comparison

Figure 53.5 shows that the errors related to the other three two-stage approximation approaches are significantly lower than those associated with APCA segmentation. The trials have shown that the ratio R method outperforms the efficiency of duration-based point identification and produces the lowest average error. As a result, when evaluating the results of Spark-based processes, the ratio R method is the best option for picking key points (Figure 53.6).

Events may have unfavorable effects on the IT infrastructure and the broader business operations. Fires in buildings, problems with central heating systems in server rooms, and unexpected equipment theft are a few examples. One successful approach is to establish procedures for recovering data during a disaster.

In such circumstances, a strategy to reduce data loss is to keep backup storage at a distant place, away from the main server equipment area. Figure 53.7 shows the comparison to information backup management.

It is essential to provide quick access to vital recovery data. Off-site data storage is a component of the described strategy, which calls for keeping backups elsewhere. Two methods are used: physically moving the data and writing it into removable drives. In the case of a failure, it is crucial to have quick access procedures in place for adequate recovery. Figure 8 shows the comparison study of off-server copy management.

The benefit of this strategy is its simplicity of organization. The difficulties in media retrieval cause the requirement to move data to preservation and the potential for media damage while in transit. It entails copying data to a different location across a network channel. Figure 53.9 shows an example of storage device management.

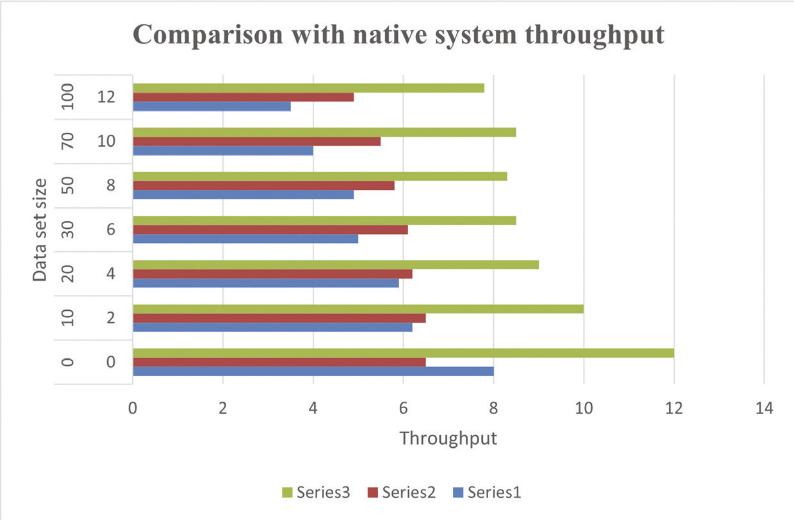

Figure 53.6 Comparison with native system throughput

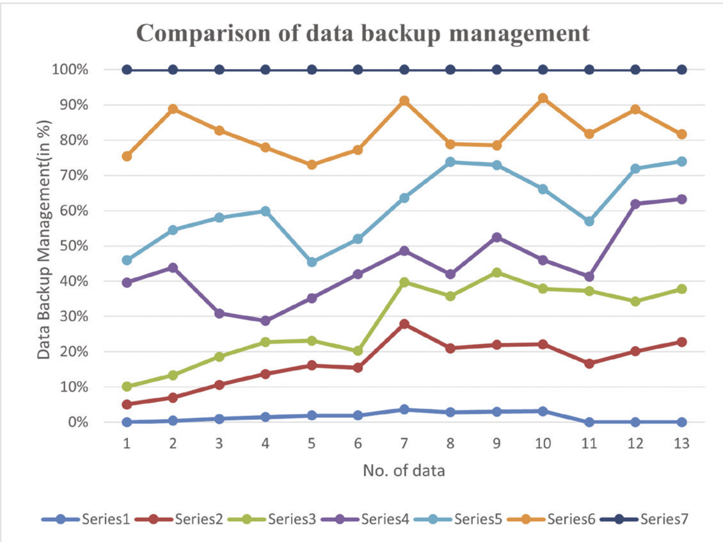

Figure 53.7 Comparison of data backup management

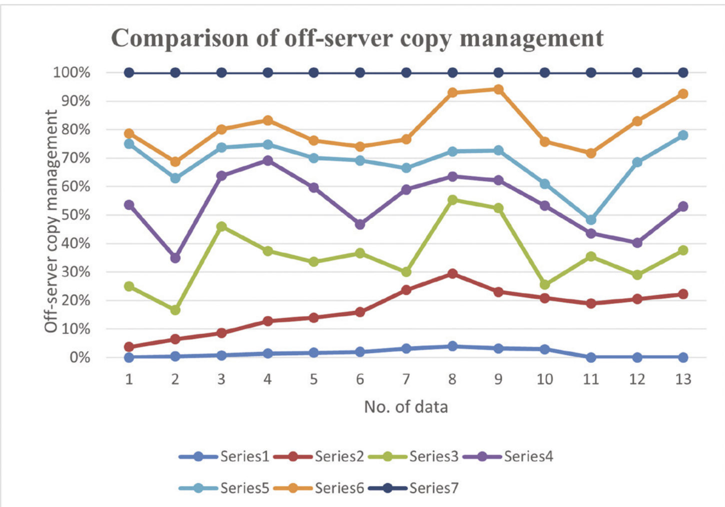

Figure 53.8 Comparison of off-server copy management

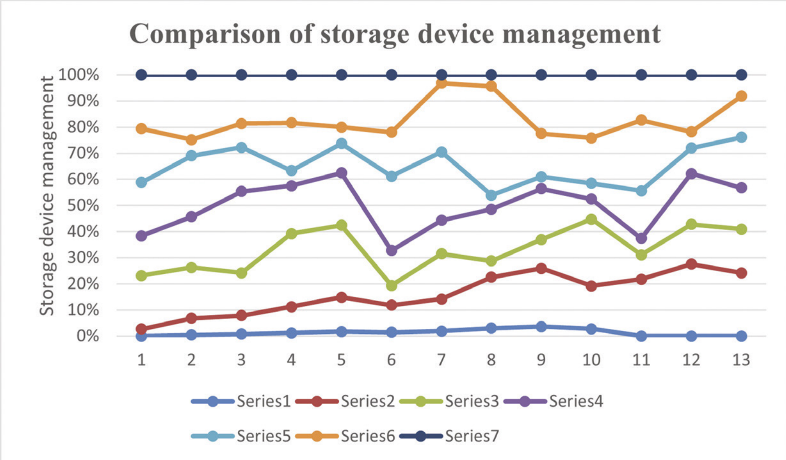

Figure 53.9 Comparison of storage device management

Conclusion

In conclusion, this study investigated cloud-based data backup and recovery techniques in depth. A thorough literature study revealed these tactics' expanding importance in response to rising cloud use and related data threats. Using a strong approach, it gathered data from numerous cloud service providers and companies and learned important lessons.

Empirical findings showed various backup techniques, with firms choosing methods following data volume, recovery goals, and budgetary restrictions. As enterprises prioritize data redundancy and disaster recovery capabilities, the choice of cloud provider has emerged as a crucial aspect. Data security, compliance and cost management were problems, but AI-driven anomaly detection and blockchain-enhanced data integrity were promising advancements. To successfully balance accessibility, security, and cutting-edge technology, it is essential to continually assess and change cloud backup and recovery procedures, as this study highlights.

References

Dalal, A. (2023). Secure cloud migration strategy (SCMS): A safe journey to the cloud. *Int. Conf. Cyber Warfare Sec.*, 18(1), 1–6. doi: 10.34190/iccws.18.1.1038.

Dajun, C., Li, L., Chang, Y., and Qiao, Z. (2021). Cloud computing storage backup and recovery strategy based on secure IoT and spark. *Mob. Inform. Sys.*, 1–13. doi: 10.1155/2021/9505249.

Durga, V. S. K., Fatima, Y., and Mailewa, A. B. (2022). Data integrity attacks in cloud computing: A review of identifying and protecting techniques. *Int. J. Res. Publ. Rev.*, 3(2), 713–720. doi: 10.55248/gengpi.2022.3.2.8.

Kiranpreet, K., Guillemin, F., and Sailhan, F. (2022). Container placement and migration strategies for cloud, fog, and edge data centers: A survey. *Int. J. Netw. Manag.*, 32(6), e2212. doi: 10.1002/nem.2212.

Iwona, K., Jackowski, A., Lichota, K., Welnicki, M., Dubnicki, C., and Iwanicki, K. (2023). InftyDedup: Scalable and cost-effective cloud tiering with deduplication. *21st USENIX Conf. File Stor. Technol.*, 23, 33–48.

Rahul, K. and Venkatesh, K. (2022). Centralized and decentralized data backup approaches. *Proc. Int. Conf. Deep Learn. Comput. Intel. ICDCI*, 2021, 687–698. doi:10.1007/978-981-16-5652-1_60.

Lai, Y. L., Rana, M. E., and Al Maatouk, Q. (2022). Critical review of design considerations in forming a cloud infrastructure for SMEs. *2022 Int. Conf. Dec. Aid Sci. Appl. (DASA)*, 1537–1543. doi: 10.1109/DASA54658.2022.9765167.

Ramesh, G., Logeshwaran, J., and Aravindarajan, V. (2023). A secured database monitoring method to improve data backup and recovery operations in cloud computing. *BOHR Int. J. Comp. Sci.*, 2(1), 1–7. doi: 10.54646/bijcs.019.

Brar, P. S., Shah, B., Singh, J., Ali, F., and Kwak, D. (2022). Using modified technology acceptance model to evaluate the adoption of a proposed IoT-based indoor disaster management software tool by rescue workers. *Sensors*, 22(5), 1866, https://doi.org/10.3390/s22051866.

Rehman, A. U., Agular, R. L., and Barraca, J. P. (2022). Fault-tolerance in the scope of cloud computing. *IEEE Acc.*, 10, 63422–63441. doi; 10.1109/ACCESS.2022.3182211.

Twana, H. S., Sharif, K. H., and Rashid, B. N. (2022). A survey of comparison different cloud database performance: SQL and NoSQL. *Passer J. Basic Appl. Sci.*, 4(1), 45–57. doi: 10.24271/psr.2022.301247.1104.

Surbhi, K. (2015). A survey on dynamic load balancing techniques in cloud computing. *Adv. Comp. Sci. Inform. Technol. (ACSIT)*, 2(7), 87–91.

Xiaojun, S., Huang, Y., Liu, Z., and Yang, Y. (2021). Reducing the service function chain backup cost over the edge and cloud by a self-adapting scheme. *IEEE Trans. Mob. Comput.*, 21(8), 2994–3008. doi: 10.1109/TMC.2020.3048885.